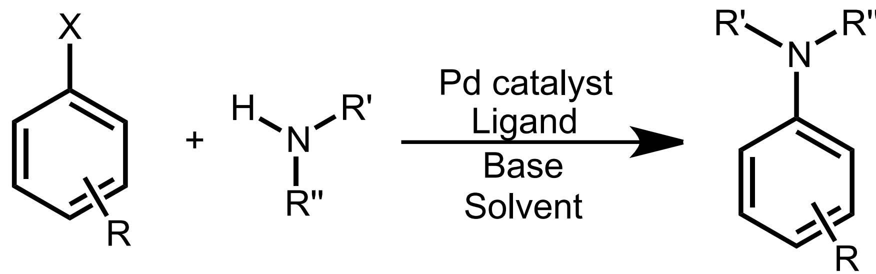
Case Study: Buchwald-Hartwig Amination

Pd-catalyzed C–N cross coupling reactions



X = Cl, Br, I, OTf
R = Almost anything
R' = aryl, alkyl, H
R'' = aryl, alkyl, H

Pd catalysts
$Pd(OAc)_2$
$Pd_2(dba)_3$
L_nPd^0
L_nPd^{II}

Base
NaOtBu
Cs_2CO_3
etc

Review articles:

- Wolfe, J. P.; Wagaw, S.; Marcoux, J.-F.; Buchwald, S. L. *Acc. Chem. Res.* **1998**, *31*, 805.
- Hartwig, J. F. *Angew. Chem. Int. Ed*. **1998**, *37*, 2046.
- Christmann, U.; Vilar, R. N. *Angew. Chem. Int. Ed.* **2005**, *44*, 366.

Reevaluation of the Mechanism of the Amination of Aryl Halides Catalyzed by BINAP-Ligated Palladium Complexes

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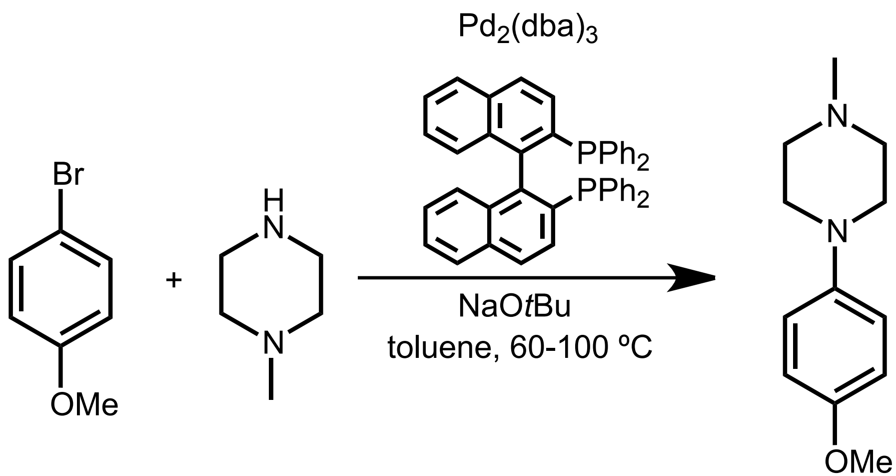
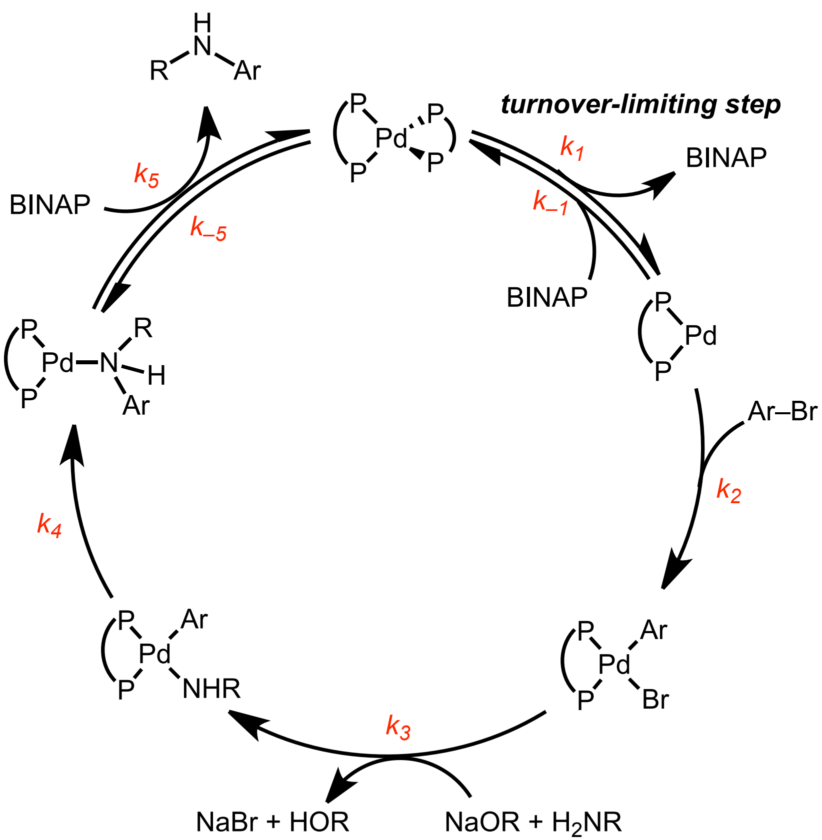
Teaming up to tackle the mechanism of one common coupling reaction:



Two Proposed Mechanisms: Both Are Incorrect!

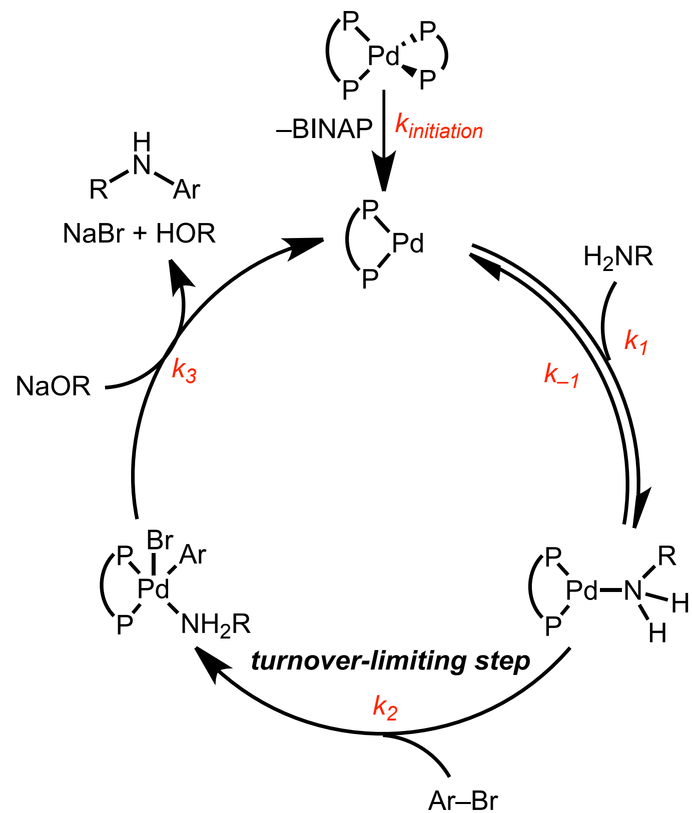
PATH A



Hartwig (2000)

Alcazar-Roman, L. M.; Hartwig, J. F.; Rheingold, A. L.; Liable-Sands, L. M.; Guzei, I. A. *J. Am. Chem. Soc.* **2000**, *122* (19), 4618–4630.

PATH B



Buchwald/Blackmond (2002)

Singh, U. K.; Strieter, E. R.; Blackmond, D. G.; Buchwald, S. L. *J. Am. Chem. Soc.* **2002**, *124* (47), 14104–14114.

Task for Today:
Determine the mechanism of the Buchwald-Hartwig reaction

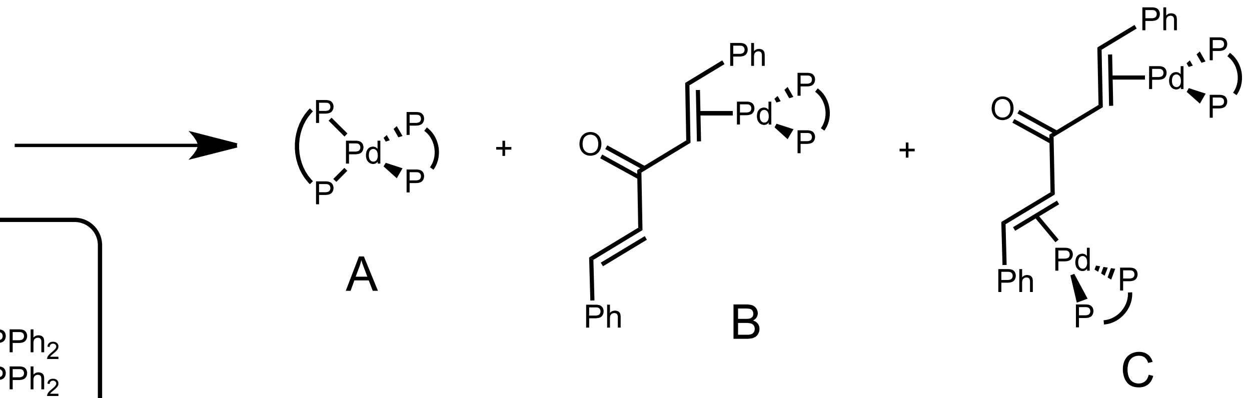
The following slides collect specific experiments from the paper and other sources.

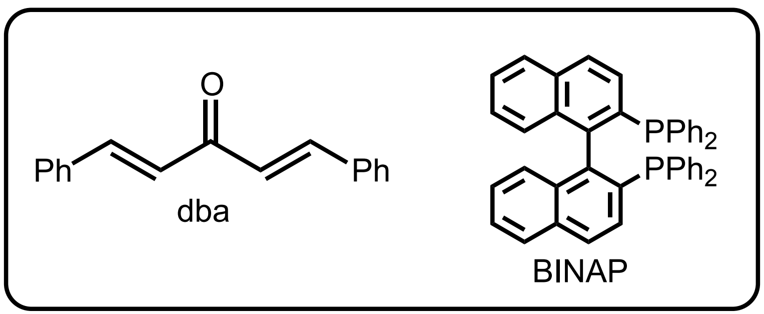
- As students suggest mechanistic probes, the instructor talks about how the experiment was done and draws the resulting data on the board.
  - The following slides can be covered in any order, depending on which experiments the students want to try first.
- The class then interprets the data and makes a mechanistic conclusion, if possible.
- At the end of the class, the data is held up against the two prior mechanisms, and alternative mechanisms are postulated.

Stoichiometric Studies

Reaction omitting aryl chloride

$Pd_2(dba)_3$ + BINAP + NaOtBu + H_2NR





- Each species was isolated and tested as a catalyst:
    - See Figure 2 of paper.
    - **A**: slight induction period
    - **B**: very slow catalyst, long induction period
    - **C**: fast catalyst, no induction period

Interpretation:
- Complex A lies off the catalytic cycle; **evidence against Path A**.
- Complex C could be a catalytic intermediate (it is catalytically competent).

Rate Law Determination

Predicted rate laws:

Path A $$rate = k_1[Pd]$$

Path B $$rate = \frac{k_1 k_2[\mathrm{Pd}][\mathrm{amine}][\mathrm{ArX}]}{k_{-1} + k_2[\mathrm{ArX}]}$$ *or* $$v = \frac{V_{\max}[\mathrm{amine}]}{K_M + [\mathrm{amine}]}$$

Order in Amine

Traditional experiment: Vary concentration of excess amine, measure ~5 pseudo-first order rate constants.

Reaction progress kinetic analysis: Perform two “different excess” experiments.

Result:
- See Figure 6 in the paper.
- Reaction kinetics are insensitive to the [amine].
- Experiment performed using RPKA.
- The identity of the amine also does not impact the rate (see Fig. 5).

Interpretation:
- Zero-order in amine.
- Rules out rate laws for Path B.
- Good for wide range of substrate scope!

Order in Aryl Halide

Traditional experiment: Vary concentration of excess ArX, measure ~5 pseudo-first order rate constants.

Reaction progress kinetic analysis: Perform two “different excess” experiments.

Result:
- See Figure 10 in the paper.
- Reaction performed using traditional methods.
- Linear relationship between k_{obs} and [ArX].

Interpretation:
- First-order in aryl halide.
- What would RPKA plots look like?

Order in Ligand

Experiment: Vary concentration of free BINAP in several reactions.

Result:

- See Figure 10 in the paper.
- Reaction performed using traditional methods.
- Linear relationship between $1/k_{obs}$ and [BINAP].

Interpretation:

- Inverse first-order in free ligand.
- Ligand inhibits the reaction

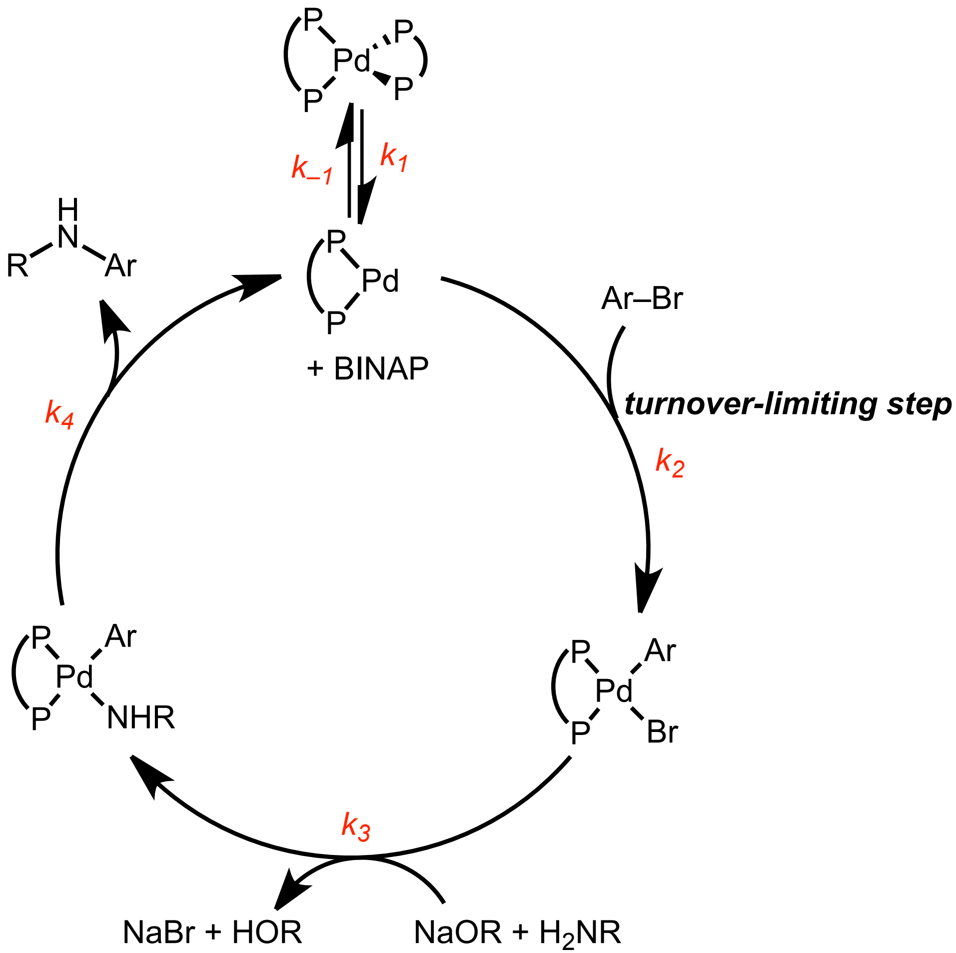
Hammett Plots

Experiment: Was not performed in the present paper.

Speculation / Predicted Result:

- Expect no or weak correlation between amine substituents and rate.
- Expect strong dependence on aryl halide substituents on rate.
- Related Pd catalysts have been studied, and typically accelerated by electron-withdrawing groups on aryl halide.

Revised Mechanism



Derive the rate law:

$$rate = \frac{k_1 k_2 [\mathrm{Pd}][\mathrm{ArX}]}{k_{-1}[\mathrm{BINAP}] + k_2[\mathrm{ArX}]} = \frac{k_1 k_2 [\mathrm{Pd}][\mathrm{ArX}]}{k_{-1}[\mathrm{BINAP}]}$$

Given in paper
(k_2 is small)

Check against observations:

- 1st order in Arx
- 0 order in amine
- Inverse 1st order in BINAP

Other Common Suggestions

Suggested experiments:

- Kinetic isotope effect study (on NH protons).
- Entropy and enthalpy of activation study.
- Study changing ligand structure.